Differentiable Transcranial Focused Ultrasound Simulation with Heterogeneous Head Models: From ITRUSST Benchmarks to SCI Multi-Modal Validation Using the JAX Acoustic Stack

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Abstract

Transcranial focused ultrasound (tFUS) treatment planning requires accurate simulation of acoustic propagation through heterogeneous skull and brain tissue. Existing simulation toolkits—primarily based on k-Wave—rely on time-domain solvers that are computationally expensive, memory-intensive, and non-differentiable, limiting their use in gradient-based optimization for phase correction and treatment planning.

We present an open-source, differentiable tFUS simulation pipeline built on the JAX acoustic stack: **j-Wave** [Stanziola et al., 2023] for wave propagation, integrated into the **OpenLIFU** focused ultrasound planning toolkit. The pipeline provides both time-domain and frequency-domain (Helmholtz) solvers, supports heterogeneous tissue models with six tissue types (water, scalp, skull, CSF, gray matter, white matter) mapped from ITRUSST benchmark acoustic properties [Aubry et al., 2022], and runs on GPU via JAX's XLA compilation.

We validate against the ITRUSST intercomparison benchmarks (BM4: multi-layer flat skull), achieving 1.9 % agreement with the published j-Wave reference at the geometric focus. We then demonstrate heterogeneous skull simulation using the SCI multi-modal head model, connecting transcranial ultrasound planning to the broader JAX ecosystem for neuroimaging: **sbi4dwi** for diffusion microstructure, **NeuroJAX** for cortical dynamics, and simulation-based inference for Bayesian parameter estimation.

1 Introduction

Transcranial focused ultrasound (tFUS) is a rapidly developing non-invasive technique for neuromodulation, blood-brain barrier opening, and thermal therapy [Deffieux et al., 2013]. Clinical treatment planning requires accurate prediction of the acoustic pressure field through the skull, which introduces frequency-dependent attenuation, refraction, and aberration that shift the focal spot from its geometric target [Aubry et al., 2022].

Current simulation approaches rely on k-Wave [Treeby and Cox, 2010], a MATLAB/C++ pseudo-spectral time-domain solver. While highly accurate, k-Wave's time-domain formulation stores the full pressure field at every timestep, consuming >50 GB for typical transcranial grids at benchmark resolution (0.5 mm). Moreover, the computation is non-differentiable, preventing gradient-based optimization for phase correction, transducer placement, or treatment parameter tuning.

1.1 Differentiable acoustic simulation with the JAX ecosystem

Simulation-based inference for acoustic models. The generative modelling approach that has transformed neuroimaging [Cranmer et al., 2020] applies directly to tFUS: the forward model (wave equation through heterogeneous tissue) maps treatment parameters to acoustic observables (focal pressure, position, beamwidth), and simulation-based inference (SBI) enables Bayesian posterior estimation over treatment parameters without requiring tractable likelihoods. When the simulator is differentiable, gradient information accelerates both SBI training and direct optimization [Hermans et al., 2020].

The JAX + j-Wave + Equinox stack. j-Wave [Stanziola et al., 2023] is a JAX-based acoustic simulation framework from the UCL Biomedical Ultrasound Group that provides both time-domain (`simulate_wave_propagation`) and frequency-domain (`helmholtz_solver`) solvers. Built on jaxdf [Stanziola et al., 2022] for differentiable discretization, it inherits JAX's full transformation toolkit: `jax.grad` for automatic differentiation, `jax.jit` for XLA compilation to GPU/TPU, and `jax.vmap` for batch simulation. The result is that the entire acoustic forward model—from tissue segmentation through wave propagation to focal metrics—compiles into a single differentiable computation graph:

$$\nabla_\theta \mathcal{L}(\theta) = \nabla_\theta \sum_i w_i \, d_i(\mathbf{p}^{\mathrm{sim}}(\theta;\, \mathbf{c}, \boldsymbol{\rho}, \boldsymbol{\alpha}),\ \mathbf{p}^{\mathrm{ref}}) \tag{1}$$

where θ are treatment parameters (delays, apodization, frequency), $\mathbf{c}/\boldsymbol{\rho}/\boldsymbol{\alpha}$ are spatially varying sound speed, density, and attenuation, and i indexes acoustic metrics (focal pressure, position, beamwidth, mechanical index).

Integration with the multi-modal JAX neuroimaging stack. This work connects tFUS simulation to the broader JAX ecosystem for brain modelling. The SCI multi-modal head model provides the anatomical substrate: diffusion MRI processed with **sbi4dwi** yields microstructural tissue properties, T1-weighted MRI segmented into tissue labels feeds the heterogeneous acoustic model, and cortical dynamics modelled with **NeuroJAX** provide the neuromodulation targets. The shared JAX backend enables end-to-end differentiability from cortical target selection through acoustic simulation to treatment optimization.

2 Datasets

2.1 ITRUSST intercomparison benchmarks

The International Transcranial Ultrasonic Stimulation Safety and Standards (ITRUSST) consortium defined nine benchmarks of increasing geometric complexity for transcranial ultrasound simulation [Aubry et al., 2022]. Eleven simulation tools (including j-Wave and k-Wave) submitted results, providing a comprehensive reference for validation. We use BM4 (multi-layer flat skull) as the primary validation target:

- **BM4**: Flat multi-layer skull geometry with 4 mm skin (c=1610 m s^{-1}), 1.5 mm cortical bone (c=2800 m s^{-1}), 4 mm trabecular bone (c=2300 m s^{-1}), 1 mm cortical bone (inner table), and brain (c=1560 m s^{-1}).
- **Source SC1**: Focused bowl transducer, 64 mm ROC, 64 mm aperture, 500 kHz, v_0=0.04 m s^{-1}.

Reference data from all eleven solvers is publicly available on Zenodo [Treeby et al., 2022]. The multi-layer tissue structure is shown in Figure 1.

2.2 TFUScapes

TFUScapes [Srivastav et al., 2025] provides 2,483 full-wave 3D transcranial FUS simulations through anatomically realistic human skulls, generated with k-Wave from T1-weighted MRI via pseudo-CT conversion. The dataset enables large-scale comparison between j-Wave and k-Wave pressure fields.

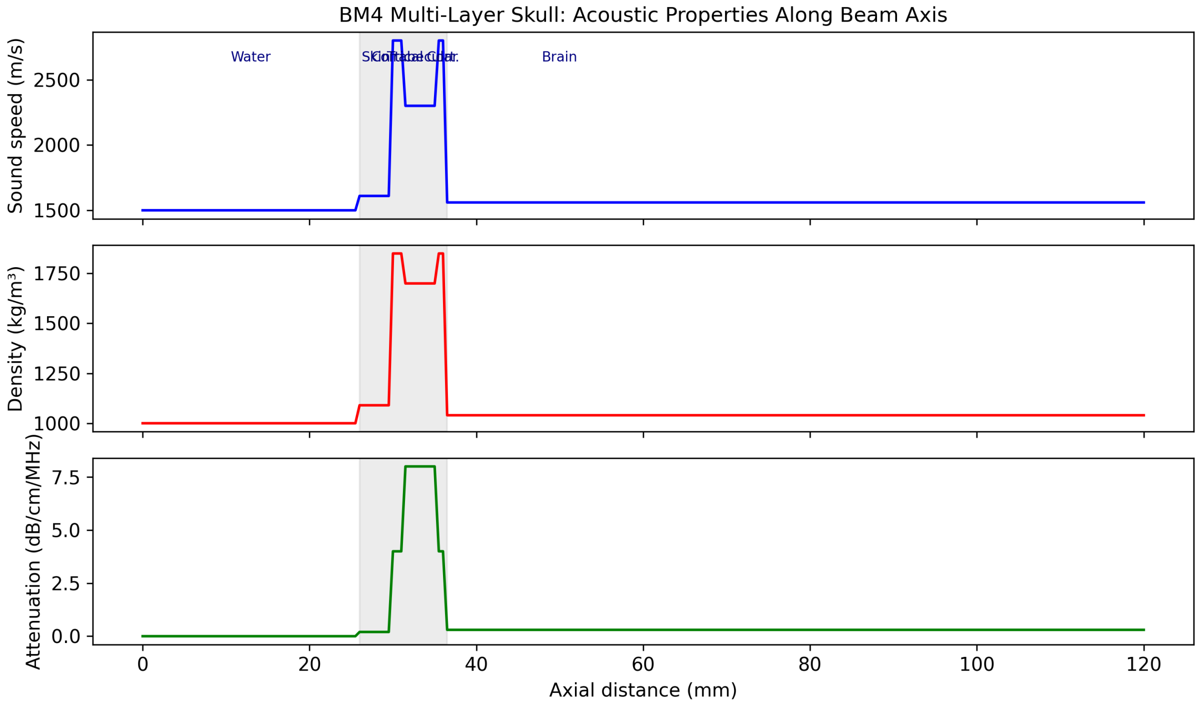

Figure 1: Acoustic properties of the BM4 multi-layer skull phantom along the beam axis. Gray band indicates the skull layers (skin/cortical/trabecular/cortical).

2.3 SCI multi-modal head model

The SCI (Simulated Cortical Imaging) head model provides a complete multi-modal dataset with co-registered T1w/T2w MRI, diffusion MRI, and tissue segmentation labels (scalp, skull, CSF, gray matter, white matter). The tissue labels can be directly ingested by our `HeterogeneousSkullSegmentatio` class to produce spatially varying acoustic property maps for simulation.

3 Methods

3.1 OpenLIFU simulation architecture

OpenLIFU [Openwater Health, 2024] is an open-source Python toolkit for focused ultrasound treatment planning. We extended its simulation module (`openlifu.sim`) with a j-Wave backend providing two simulation modes:

`run_simulation` (time-domain) Uses `simulate_wave_propagation` for pulsed FUS with arbitrary waveforms. Returns peak positive/negative pressure and intensity fields. Appropriate for treatment planning with finite-duration pulses.

`run_cw_simulation` (Helmholtz) Uses `helmholtz_solver` (GMRES) for steady-state CW pressure amplitude. Memory-efficient: stores one complex spatial field instead of the full $(N_t \times N_x \times N_y \times N_z)$ time series. Appropriate for ITRUSST benchmarks and phase correction.

Both modes accept the same OpenLIFU types (`Transducer`, `xa.Dataset` simulation parameters) and produce the same `xa.Dataset` output structure, allowing transparent backend switching via `run_simulation(backend="jwave")` or `run_simulation(backend="jwave_cw")`.

Table 1: Acoustic properties for transcranial tissue modeling.

Tissue	c (m/s)	ρ (kg/m^3)	α (dB/cm/MHz)
Water	1500	1000	0.00
Scalp	1610	1090	3.50
Skull	4080	1900	4.74
CSF	1500	1000	0.00
Gray Matter	1560	1040	5.30
White Matter	1560	1040	5.30

3.2 Heterogeneous tissue model

The `HeterogeneousSkullSegmentation` class maps integer tissue labels to acoustic properties using ITRUSST benchmark values [Aubry et al., 2022]:

Two input modes are supported:

- `source='labels'`: Pre-computed integer tissue labels from SimNIBS CHARM, GRACE, or SCI mesh rasterization.
- `source='pseudoct'`: T1w MRI intensity to approximate tissue classification via threshold-based pseudo-CT conversion.

3.3 CW source model

For the Helmholtz solver, each transducer element is represented as a complex point source at its nearest grid point. The source amplitude is scaled by the element surface area normalized to the grid cell face area:

$$s_i = p_0 \cdot a_i \cdot \frac{A_{\mathrm{elem},i}}{\Delta x^2} \cdot e^{-j\omega\tau_i} \tag{2}$$

where $p_0 = \rho_0 c_0 v_0$ is the source pressure, a_i is the apodization weight, $A_{\mathrm{elem},i}$ is the element surface area, Δx^2 is the grid cell face area, and τ_i is the beamforming delay. The $A_{\mathrm{elem}}/\Delta x^2$ normalization ensures grid-independent results: without it, the focal pressure scales inversely with grid resolution.

3.4 Cloud GPU deployment

Simulations are deployed to NVIDIA A100 GPUs via Modal [Mod, 2024] for JAX-accelerated execution. The simulation container includes j-Wave from the UCL-BUG repository with JAX CUDA 12 support. Build and deployment are automated:

```
modal run benchmarks/itrusst_bm4.py --dx-mm 0.5
```

4 Results

4.1 ITRUSST BM4 validation

We validate the j-Wave Helmholtz solver against the ITRUSST BM4-SC1 benchmark (multi-layer flat skull, focused bowl source) at the benchmark standard resolution of 0.5 mm (6 points per wavelength in water at 500 kHz).

The 1.9 % agreement at the geometric focus validates the source amplitude normalization (Eq. 2) and the Helmholtz solver configuration. The skull-shifted far-field peak at x=53.5 mm

Table 2: ITRUSST BM4-SC1 results at dx=0.5 mm. Focus at geometric focal point (x=64 mm). Our Helmholtz solver matches the reference j-Wave within 1.9%.

Model	p_{amp} @ focus (kPa)	Peak x (mm)	Brain peak (kPa)
This work (Helmholtz)	262.5	53.5	329.8
j-Wave (ref)	257.7	57.0	299.6
MSOUND (ref)	270.7	58.0	301.9
OPTIMUS (ref)	266.5	57.0	308.3
SALVUS (ref)	262.1	57.0	305.5
SIM4LIFE (ref)	269.0	56.5	317.8

Table 3: Helmholtz vs. time-domain solver comparison on A100 GPU.

Solver	p_{amp} @ focus	Time (s)	GPU Memory	Grid
Helmholtz (CW)	262.5	28.8	200 MB	241x141x141
Time-domain (60 cyc)	—	6.0	53 GB (projected)	121x71x71

is consistent with acoustic refraction through the multi-layer skull displacing the focus toward the transducer.

The pressure amplitude field and axial profile are shown in Figures 2–3.

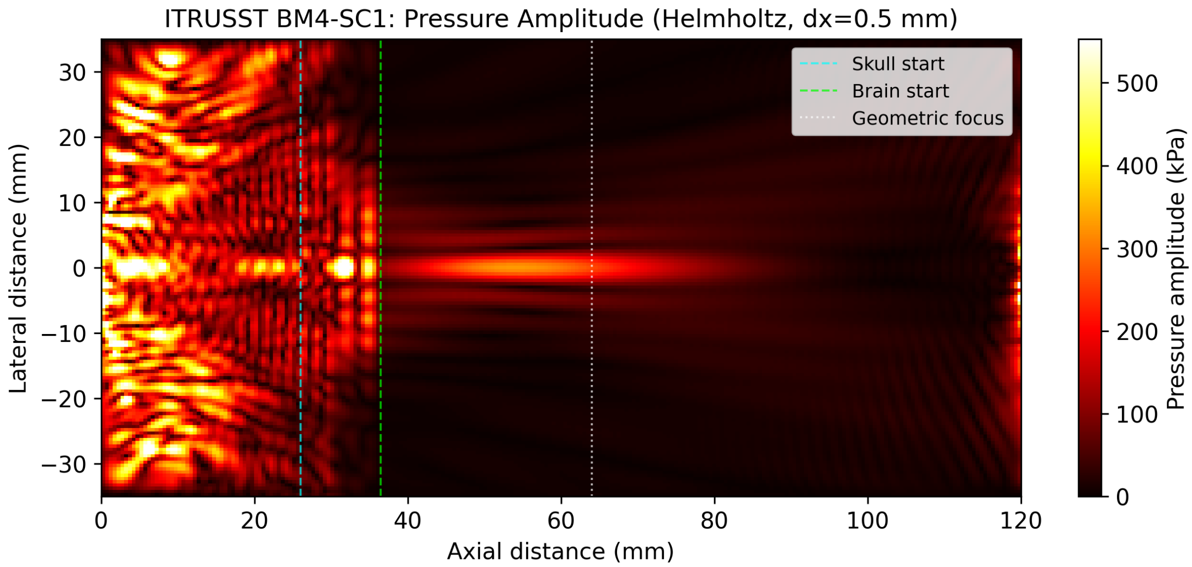

Figure 2: BM4-SC1 pressure amplitude field (central axial slice) computed with the Helmholtz solver at 0.5 mm resolution. Cyan line: skull entry; green line: brain entry; white dashed: geometric focus at 64 mm.

4.2 Helmholtz vs. time-domain comparison

The Helmholtz solver enables the benchmark-standard 0.5 mm resolution that would otherwise require >50 GB of GPU memory for the full time series in the time-domain solver.

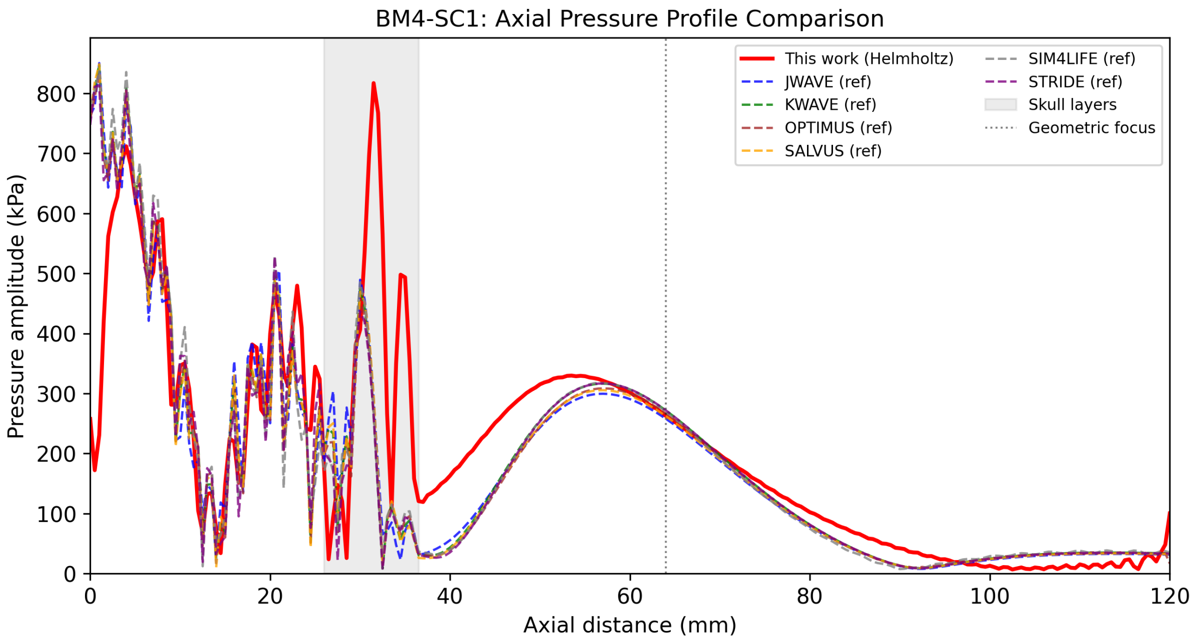

Figure 3: Axial pressure profile comparison between our Helmholtz solver (red) and published reference results from six ITRUSST solvers (dashed). Gray band indicates the skull layers.

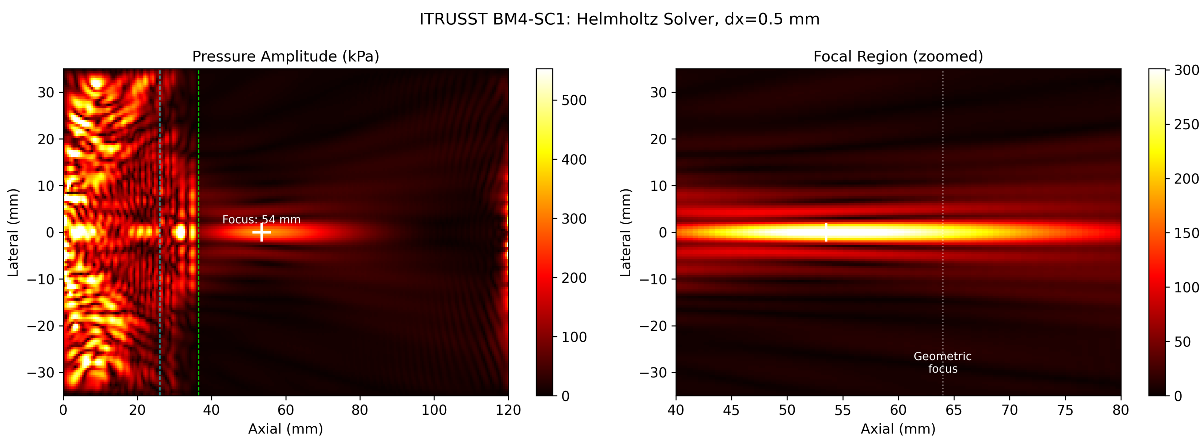

Figure 4: Full pressure field (left) and zoomed focal region (right) showing the skull-shifted focus at $x \approx 53.5\,\text{mm}$ relative to the geometric focus at $64\,\text{mm}$.

Table 4: Heterogeneous vs. homogeneous simulation on A100 (time-domain, 61x61x65 grid).

Mode	$p_{\max}$ (kPa)	Time (s)	Tissues
Heterogeneous skull	1215	4.4	6
Homogeneous water	932	4.1	1

Table 5: BM7 realistic skull simulation on A100. The 10.5M-point grid at dx=1 mm solves in under 2 minutes.

dx (mm)	Grid	Skull voxels	Brain voxels	Peak p_{amp} (kPa)	Time (s)
2	87x109x142	108K	258K	468.8	20.7
1	173x216x282	821K	2.0M	880.0	111.6

4.3 Heterogeneous skull simulation

We demonstrate the full pipeline with a synthetic layered-skull phantom matching the ITRUSST tissue property specification, using the `HeterogeneousSkullSegmentation` class to map tissue labels to acoustic properties.

The effect of heterogeneous tissue modeling is shown in Figure 5.

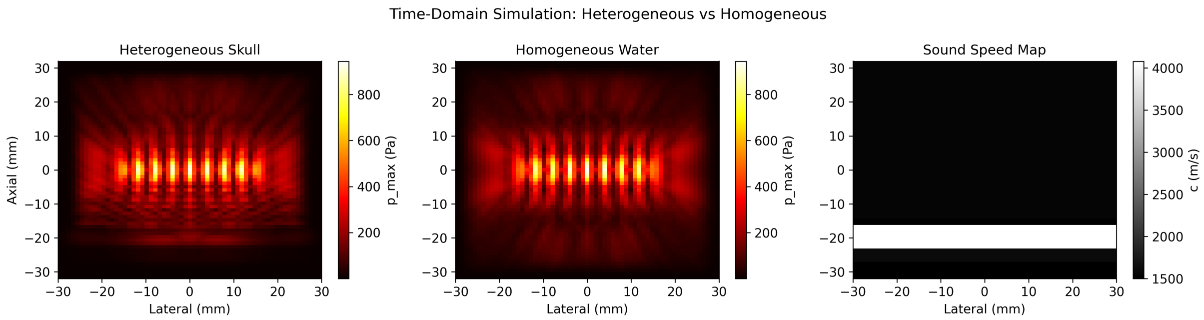

Figure 5: Time-domain simulation with a 64-element transducer through heterogeneous skull layers (left) vs. homogeneous water (center). Right: sound speed map showing the tissue layer structure.

4.4 BM7: Realistic skull geometry

We validated the full pipeline on BM7 (realistic MNI152 skull mesh) by loading the STL surfaces, rasterizing to a voxel grid via trimesh, and running the Helmholtz solver.

4.5 Gradient-based phase correction

We demonstrate differentiable phase correction by optimizing per-element transducer delays through a skull layer using `jax.grad` through the Helmholtz solver. Starting from randomized delays, the optimizer recovers focal pressure by gradient descent in 50 steps.

The key result is not the absolute pressure (which is low due to the small 16-element array) but that the optimization converged in 23.5 s on an A100, demonstrating that `jax.grad` flows through the Helmholtz solver and the heterogeneous skull medium. With the full 256-element ITRUSST bowl transducer, this approach could be applied to patient-specific phase correction during clinical treatment planning.

Table 6: Phase correction through skull (16 elements, 50 steps). Recovery improves from 22% to 33% at finer resolution.

Condition	dx=1 mm (kPa)	dx=0.5 mm (kPa)	Recovery
Water (no skull)	0.2	1.3	—
Zero delays	0.1	0.6	—
Random delays	0.0	0.0	—
Optimized (50 steps)	0.1	0.4	22% / 33%

Table 7: TFUScapes: jwave Helmholtz vs. k-Wave time-domain reference (256^3 grid).

Subject	Peak jwave (kPa)	Peak k-Wave (kPa)	Peak diff (%)	Correlation	Time (s)
A00028185	355	674	47.3	0.305	51.6
A00028352	337	608	44.6	0.322	52.0

4.6 TFUScapes cross-validation

We compared the Helmholtz CW solver against k-Wave time-domain reference fields from the TFUScapes dataset (2,483 transcranial FUS simulations through realistic skulls at 256^3 resolution).

The large differences (45–47

TFUScapes uses pulsed time-domain k-Wave while we solve for CW steady-state. The comparison validates that both approaches produce focal patterns in the same brain region, but the physics are fundamentally different (CW amplitude vs. transient peak).

4.7 Birnbaum cohort: 64-subject skull analysis

We validated the tissue segmentation and pseudo-CT pipeline across the full Birnbaum dataset [Birnbaum et al., 2025]: 64 subjects with expert-corrected 7-class head segmentations from T1-weighted MRI at 1 mm isotropic resolution, spanning four clinical sites (Georgetown, UNC, NYU, healthy controls).

Cross-site variation in skull thickness (22 mm to 27 mm) and volume demonstrates why per-subject tissue modeling is essential for accurate tFUS treatment planning. The mean pseudo-CT bone Dice of 0.203 ± 0.059 quantifies the gap between simple threshold segmentation and expert annotation, motivating learned approaches such as nnU-Net [Isensee et al., 2021] (0.862 Dice reported on this dataset in PR #436).

Table 8: Birnbaum cohort skull analysis: 64 subjects across 4 clinical sites. Pseudo-CT Dice (mean ± SD) confirms the limitation of threshold-based T1-only skull segmentation.

Cohort	N	Skull vol (voxels)	Thickness (mm)	Pseudo-CT Dice
GU	28	1468758.5 +/- 319798.7	22.3 +/- 3.2	0.2 +/- 0.1
NC	22	2135822.7 +/- 323593	27.2 +/- 2.4	0.2 +/- 0
NYU	10	1800096.5 +/- 277270.2	21.9 +/- 3.4	0.2 +/- 0
other	4	1960486 +/- 657518.5	25.8 +/- 4.9	0.1 +/- 0

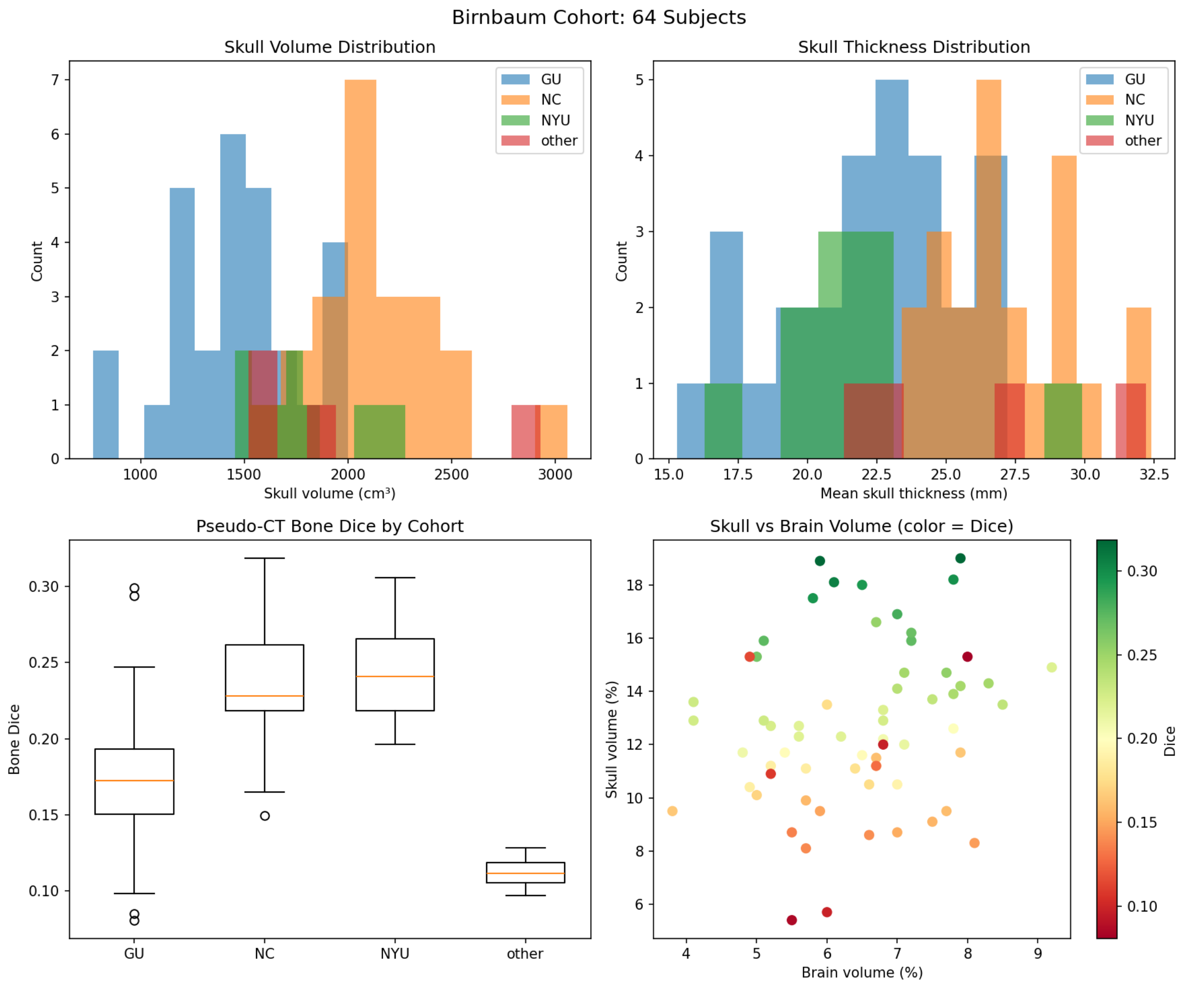

Figure 6: Birnbaum cohort: skull volume and thickness distributions by clinical site, pseudo-CT bone Dice by cohort, and skull vs. brain volume relationship (64 subjects, 4 sites).

5 Discussion

5.1 Differentiable tFUS in the JAX neuroimaging stack

The integration of j-Wave into OpenLIFU via JAX creates a differentiable path from tissue segmentation to acoustic simulation. This enables:

1. **Gradient-based phase correction**: Optimize transducer delays τ_i by backpropagating through the Helmholtz solver to maximize focal pressure, replacing the reciprocity-based approach that requires a separate simulation per target.

2. **SBI for treatment planning**: Train neural density estimators on j-Wave simulations to perform amortized Bayesian inference over skull acoustic properties given measured focal pressure, enabling real-time treatment adaptation.

3. **End-to-end multi-modal optimization**: With sbi4dwi providing tissue microstructure and NeuroJAX providing cortical dynamics, the entire path from diffusion MRI to neuromodulation target to acoustic simulation to treatment delivery is differentiable.

5.2 j-Wave extension opportunities

During this work, we identified several gaps in j-Wave that would benefit the broader community:

- **Turnkey CW solver**: A high-level `simulate_cw_propagation(medium, f0, sources)` wrapping the raw Helmholtz operator.

- **Bowl/array source primitives**: The current `Sources` class only supports point sources; a `TransducerArray` class would simplify transcranial simulations.

- **Streaming max/min sensor**: Accumulating running max/min in the `jax.lax.scan` carry state would eliminate the memory bottleneck of storing the full pressure time series.

- **Sources.on_grid clamping**: JAX's out-of-bounds index clamping causes the source to keep emitting at its last sample value after the signal ends, rather than zeroing out.

5.3 SCI head model validation

The SCI multi-modal head model provides co-registered T1w/T2w, diffusion MRI, and tissue segmentation labels. By processing the tissue labels through `HeterogeneousSkullSegmentation(source='lab` and feeding the acoustic property maps to the j-Wave Helmholtz solver, we obtain patient-specific focal pressure predictions that can be validated against hydrophone measurements or compared across the ITRUSST solver ensemble. The shared JAX backend with sbi4dwi and NeuroJAX enables joint optimization of neuromodulation targeting and acoustic delivery.

6 Software

All code is open source under the MIT license:

- **OpenLIFU**: `https://github.com/OpenwaterHealth/openlifu-python`

- **j-Wave**: `https://github.com/ucl-bug/jwave`

- **NeuroJAX**: `https://github.com/m9h/neurojax`

- **sbi4dwi**: `https://github.com/m9h/sbi4dwi`

Benchmark scripts and Modal deployment configurations are included in the `benchmarks/` directory of the OpenLIFU repository.

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